CHAPTER 6

# Differentiation, variation and optimisation

**Chapter path**

**Objectives.** Calculate a derivative on its domain; study variation and extrema; compare curves; optimise; use symmetries.
**Prerequisites.** Derivative at a point, algebraic calculation, sign of a quadratic, intervals.
**Yearly reference point.** 1–10 December: 8 hours, consolidation on the 15th and 17th

## Diagnostic: six questions, 10 minutes

Answer without consulting the lesson. Mark the questions for which help is needed. This diagnostic is not graded.

1. Derivative of $x^2$ at 3?
2. Domain of $\dfrac{1}{x-2}$?
3. Sign of $(x-1)(x+2)$?
4. Expand $(x^2+1)(x-2)$.
5. For $f(x)=x^2$, compare $f(-x)$ and $f(x)$.
6. If $f(1)=2$, $f\prime(1)=0$, what is the equation of the tangent?

**After the diagnostic.** Revisit the flagged topics with the teacher; redo a similar question, then explain the rule used.

**Speaking and logic.** Present a variation table, distinguishing the sign of f from the sign of its derivative. Explain why the endpoints of an optimisation problem must be examined.

## Investigate before formalising

Can an increasing function have a horizontal tangent? Look for an example; can it have a strictly negative derivative at an interior point?

**Organisation.** Individual investigation 5 min, comparison in pairs 5 min, class discussion 5 min. Keep a record of one attempt, even an unfinished one. A conjecture must then be justified.

## 6.1 Derivative function and standard functions

**Definition and results**

A function is differentiable on an interval when it is differentiable at each of its points; its derivative associates with x the number $f\prime(x)$. The standard derivatives are: constant $\mapsto 0$; $x \mapsto 1$; $x^2 \mapsto 2x$; $x^3 \mapsto 3x^2$; $\frac{1}{x} \mapsto -\frac{1}{x^2}$ for $x \neq 0$; $\sqrt{x} \mapsto \frac{1}{2\sqrt{x}}$ for $x > 0$. For $n \in \mathbb{Z}$, $(x^n)\prime = nx^{n-1}$ on its domain of differentiability; for $n = 0$, the derivative is 0 directly.

**Status and justification.** Square: the rate is $2a + h$. Reciprocal: for $a \neq 0$, $h \neq 0$ and $a + h \neq 0$, $\frac{[\frac{1}{a+h} - \frac{1}{a}]}{h} = -\frac{1}{[a(a+h)]}$, which tends to $-\frac{1}{a^2}$. The other formulas are accepted here.

**Worked example.** For $f(x) = x^{-3}$, the domain is $\mathbb{R} \setminus \{0\}$ and $f\prime(x) = -3x^{-4} = -\frac{3}{x^4}$. For $g(x) = \sqrt{x}$, the function exists at 0, its standard derivative only for $x > 0$.

**Watch out**

The domain of differentiability may be smaller than the domain of definition.

**Immediate application.** Differentiate $x^5$, $x^{-2}$ and $\sqrt{x}$, stating the domains.

## 6.2 Sum, product, reciprocal and quotient

**Definition and results**

For differentiable u and v, $(u + v)\prime = u\prime + v\prime$, $(\lambda u)\prime = \lambda u\prime$, $(uv)\prime = u\prime v + uv\prime$. Where $v \neq 0$, $(\frac{1}{v})\prime = -\frac{v\prime}{v^2}$ and $(\frac{u}{v})\prime = \frac{u\prime v - uv\prime}{v^2}$.

**Status and justification.** For the product, add and subtract $u(a+h)v(a)$ in the numerator. The rate becomes $\frac{u(a+h)[v(a+h) - v(a)]}{h} + \frac{v(a)[u(a+h) - u(a)]}{h}$. Since a differentiable function is continuous (accepted), we obtain $u(a)v\prime(a) + v(a)u\prime(a)$. The other rules are accepted.

**Worked example.** For $f(x) = \frac{x^2+1}{x-1}$, $x \neq 1$, $f\prime(x) = \frac{[2x(x-1) - (x^2+1)]}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}$. Keeping the brackets avoids changing the sign of a single term instead of the whole numerator.

**Watch out**

The derivative of a product is not the product of the derivatives; the derivative of a quotient is not their quotient.

**Immediate application.** Differentiate $(x^2+1)(3x-2)$, then $\dfrac{1}{x^2+1}$.

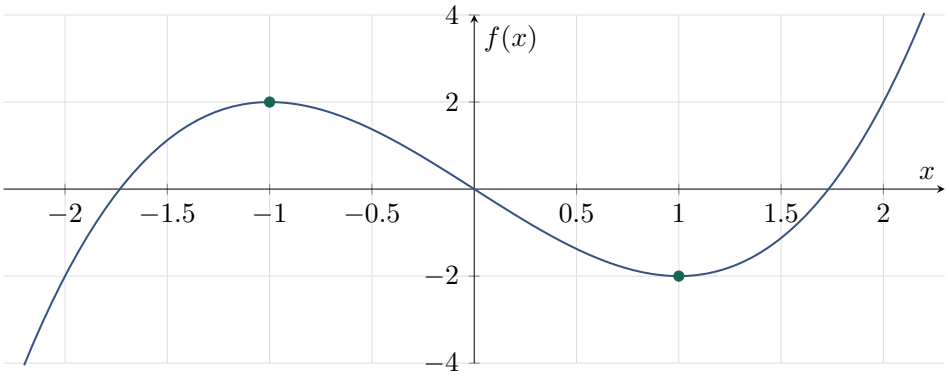
## 6.3 Sign of the derivative and extrema

**Definition and results**

On an interval I, if $f' \geq 0$, f is increasing; if $f' \leq 0$, it is decreasing; if $f' = 0$ everywhere, it is constant. A strictly positive derivative implies strict increase. If f has a local extremum at an interior point where it is differentiable, its derivative is zero there. The converse is false.

**Status and justification.** Variation theorems accepted. For an interior minimum at a, the rates on the right are positive or zero, those on the left negative or zero; if they converge to the same real number, it is 0. Same argument for a maximum.

**Worked example.** $f(x) = x^3 - 3x$: $f'(x) = 3(x-1)(x+1)$. f increases on $]-\infty;-1]$, decreases on $[-1;1]$, increases on $[1;+\infty[$; local maximum $f(-1) = 2$, local minimum $f(1) = -2$. Neither of these extrema is global on $\mathbb{R}$.




$f(x) = x^3 - 3x$ : a local maximum at $-1$ and a local minimum at 1.

**Watch out**

The theorem applies on an interval. A function can decrease on two separate intervals without decreasing on their union.

**Immediate application.** Study $f(x) = x^2 - 4x + 1$ on $[0;3]$ and give its extrema on this interval.

## 6.4 Parity, inequalities and comparison

**Definition and results**

On a domain symmetric about 0, f is even if $f(-x) = f(x)$ and odd if $f(-x) = -f(x)$. The curve of an even function is symmetric about the y-axis; that of an odd function about the origin. To compare f and g, we study the sign of $f - g$, possibly with its derivative.

**Status and justification.** The symmetries follow from the coordinates of the image points. For an inequality, a study of variation may give a minimum of the difference, then its sign; the derivative does not directly replace this difference.

**Worked example.** For $x > 0$, $h(x) = x + \dfrac{1}{x} - 2$ has derivative $\dfrac{x^2 - 1}{x^2}$. It decreases up to 1 then increases; $h(1) = 0$, so $x + \dfrac{1}{x} \geq 2$. Equality is obtained only at 1.

**Watch out**

Check the domain before parity. The sign of $f' - g'$ tells us about the variation of $f - g$, not directly about its sign.

**Immediate application.** Study the parity of $x^4 + x^2$ and $x^3 - x$; prove $x^2 + 1 \geq 2x$.

## 6.5 Optimisation and Newton's method

**Definition and results**

Optimising means defining the variable and its domain, expressing the quantity to optimise, then comparing all the candidate values, including the endpoints. Newton's method uses the tangent to approximate a solution of $f(x) = 0$: $x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$ when $f'(x_n) \neq 0$. It is used only in favourable cases.

**Status and justification.** Newton's formula comes from the intersection of $y = f(x_n) + f'(x_n)(x - x_n)$ with the horizontal axis. Convergence is not guaranteed in general; it is observed and checked in the case chosen.

**Worked example.** For $f(x) = x^2 - 2$ with $x_0 = 1$, Newton gives $x_1 = \dfrac{3}{2}$, $x_2 = \dfrac{17}{12}$. The squares are 2.25 then about 2.00694: we approach $\sqrt{2}$. The program must check for a zero derivative and a maximum number of steps.

**Watch out**

A numerical value produced by an algorithm is not a proof of a minimum or of convergence.

**Immediate application.** A rectangle of perimeter 12 m has width x. Determine the maximum area by differentiation.

## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 6.1 – A collection of derivatives

Consider the functions

$$f(x) = 3x^4 - 2x + 7, \quad g(x) = (x^2 + 2)(x - 1), \quad h(x) = \frac{2x + 1}{x + 3}.$$

1. State the domain of definition of each of these functions.
2. Calculate their derivatives, naming the rule used each time. For $g$, first keep the two terms of the product rule; for $h$, write the numerator before simplifying it.
3. Check the result for $g$ by expanding its expression before differentiating.

### Exercise 6.2 – A complete table

The function $f$ is defined on $[0; 4]$ by $f(x) = x^3 - 6x^2 + 9x + 1$. We look for its minimum and maximum values on this interval.

1. Calculate and factorise $f'(x)$, then study its sign on $[0; 4]$.
2. Draw the variation table of $f$, giving the values at the endpoints and at the points where the derivative vanishes.
3. Give the minimum and the maximum of $f$ on the interval, as well as all the x-coordinates where they are reached.

### Exercise 6.3 – A quotient

For every strictly positive real number $x$, let $f(x) = x + \dfrac{4}{x}$.

1. Study the variation of $f$ on $]0; +\infty[$. Present the factors that determine the sign of the derivative.
2. Deduce the smallest value of $x + \dfrac{4}{x}$, then write an inequality valid for all $x > 0$. State exactly its equality case.

### Exercise 6.4 – Comparing two curves

The functions $f(x) = x^3$ and $g(x) = x$ are defined on $\mathbb{R}$. Let $\mathcal{C}_f$ and $\mathcal{C}_g$ be their curves in the same frame.

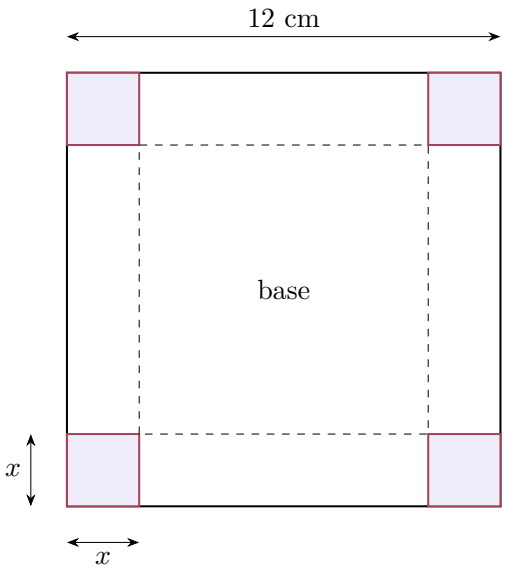
1. Determine the points of intersection of these two curves.

2. Study their relative position on $\mathbb{R}$ using a suitable method. Express your result as the intervals where $\mathcal{C}_f$ is above or below $\mathcal{C}_g$.

## Exercise 6.5 – A box

A box without a lid is made from a square sheet of side 12 cm. A square of side $x$ cm is cut from each of the four corners, then the remaining edges are folded up. The thickness of the sheet is neglected.

1. Draw the net and express the height and the two dimensions of the base of the box. State the values of $x$ that give a box of nonzero volume.
2. Express the volume $V(x)$ in cubic centimetres. Determine the dimensions that make this volume maximal.
3. Check the values at the two endpoints of the domain and give the exact maximum volume.




The four coloured regions are removed; the dotted lines show the folds. The drawing uses an illustrative value of x and does not give the optimum.

## Exercise 6.6 – Stationary without an extremum

A student claims: "If the derivative vanishes at a point, the function necessarily has a local extremum at that point." We study the function $f(x) = x^3$ to test this claim.

1. Calculate $f\prime(0)$.
2. Compare with $f(0)$ the values of $f(x)$ for negative then positive real numbers $x$, as close to 0 as we like.
3. Conclude about the existence of a local extremum at 0 and about the claim proposed.

## Exercise 6.7 – Comparing strategies

We wish to prove that, for every real number $x$, $x^2 - 6x + 12 \geq 3$.

1. Give a proof by making a square appear.

2. Give a second proof by studying the variation of a function.

3. State the equality case. Compare the two methods: which requires the least calculation here?

### Exercise 6.8 – Delayed review

The reciprocal function $f(x) = \dfrac{1}{x}$ has domain $D = \mathbb{R} \setminus \{0\}$. A student calculates $f\prime(x) = -\dfrac{1}{x^2} < 0$ and concludes that $f$ is decreasing on all of $D$.

1. Identify the missing hypothesis in the application of the theorem linking the sign of the derivative and variation.

2. State the two intervals on which the conclusion is correct.

3. Using the numbers $-1$ and $1$, construct a counterexample to the conclusion on all of $D$.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 12 p. 156, questions 1 and 2.** Calculate the difference, study its sign and translate this into the relative position of the curves.

**CQFD, exercise 16 p. 156, questions f and g.** For each function, give an interval on which it is both negative and increasing; justify the two properties separately.

## MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $(x^3)'$ equals
   - A. $x^2$
   - B. $3x^2$
   - C. $3x$
   - D. $\dfrac{x^4}{4}$
2. $\left(\dfrac{1}{x}\right)'$ equals
   - A. $-\dfrac{1}{x^2}$
   - B. $\dfrac{1}{x^2}$
   - C. $-\dfrac{1}{x}$
   - D. $1$
3. $(uv)'$ equals
   - A. $u'v'$
   - B. $u' + v'$
   - C. $uv'$
   - D. $u'v + uv'$
4. $f' > 0$ on an interval implies
   - A. f positive
   - B. f negative
   - C. f strictly increasing
   - D. f constant
5. $\dfrac{x^2+1}{x-1}$ is defined if
   - A. $x > 0$
   - B. $x \neq 1$
   - C. $x \neq -1$
   - D. $x \neq 0$

## MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $x^3 - x$ is
   - A. even
   - B. constant
   - C. odd
   - D. neither even nor odd
2. $f\prime(a) = 0$ necessarily implies
   - A. a horizontal tangent
   - B. a maximum
   - C. a minimum
   - D. f constant
3. Derivative of $\sqrt{x}$:
   - A. $2\sqrt{x}$
   - B. $\dfrac{1}{\sqrt{x}}$
   - C. $\dfrac{\sqrt{x}}{2}$
   - D. $\dfrac{1}{2\sqrt{x}}$, $x > 0$
4. Minimum of $x + \dfrac{1}{x}$ on $]0;+\infty[$:
   - A. 0
   - B. 2
   - C. 1
   - D. none
5. Newton's method for $x^2 - 2$ is not defined with $x_0 =$
   - A. 1
   - B. 2
   - C. 0
   - D. $-1$

### True or false? Justify.

1. The derivative of a positive function is positive.
2. An interior extremum of a differentiable function gives a zero derivative.
3. The product of two increasing functions is increasing.

**Self-assessment**

I differentiate on the right domain; I simplify without losing the forbidden values; I distinguish local from global extremum; I sometimes choose a factorisation rather than a derivative.
For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.